Olympiad Test Paper — Mathematics (Class 7)

Chapter 7: A Tale of Three Intersecting Lines

Subject	Mathematics (NCERT Class 7)
Chapter	7 — A Tale of Three Intersecting Lines (Triangles)
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Use the angle-sum (180°), the exterior-angle theorem, and the triangle inequality as your main tools, and sketch a quick figure where one is described in words. Watch the classic traps: a triangle is "acute" only when ALL three angles are under 90°, two sides must SUM to MORE than the third (equal is not enough), and an altitude is a perpendicular, not a side. Do not write on this paper — mark answers on your answer sheet.

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1. Which of the following correctly names the parts of triangle ΔABC?

(A) Vertices A, B, C; sides AB, BC, CA; angles ∠A, ∠B, ∠C
(B) Vertices AB, BC, CA; sides A, B, C; angles ∠A, ∠B, ∠C
(C) Vertices A, B, C; sides ∠A, ∠B, ∠C; angles AB, BC, CA
(D) Sides A, B, C; vertices AB, BC, CA; angles ΔABC

2. Three points P, Q, R are marked so that PQ = 4 cm, QR = 3 cm and PR = 7 cm. What can you say about them?

(A) They form a scalene triangle
(B) They are collinear, so no triangle is formed
(C) They form an isosceles triangle
(D) They form an obtuse triangle

3. Two angles of a triangle are 48° and 67°. The third angle is:

(A) 75°
(B) 115°
(C) 65°
(D) 55°

4. Can a triangle have sides 5 cm, 8 cm and 13 cm?

(A) Yes, because all three sides are different(B) Yes, because 8 + 13 > 5
(C) No, because the sides are too long
(D) No, because 5 + 8 = 13 is not greater than 13

5. Two sides of a triangle measure 7 cm and 10 cm. The third side x must satisfy:

(A) $7 < x < 10$ (B) $3 < x < 17$
(C) $x < 17$ (D) $3 \leq x \leq 17$

6. A triangle has angles 40°, 60° and 80°. Classified by its angles it is:
(A) Acute (B) Right
(C) Obtuse (D) Equiangular and right

7. An exterior angle of a triangle measures 120°. One of the two interior angles not adjacent to it is 70°. The other one is:
(A) 60° (B) 110°
(C) 70° (D) 50°

8. In ΔABC, $\angle A = 90°$ and $\angle B = 90°$ is claimed. Such a triangle:
(A) Is a right isosceles triangle (B) Is obtuse
(C) Has its third angle equal to 0°, so it cannot exist
(D) Has third angle 90° too

9. A triangle has angles 90°, 45° and 45°. By sides AND by angles it is:
(A) Right-angled and isosceles (B) Acute and isosceles
(C) Right-angled and scalene (D) Obtuse and isosceles

10. Which set of three lengths (in cm) CANNOT form a triangle?
(A) 6, 6, 6 (B) 4, 9, 6
(C) 5, 12, 13 (D) 2, 3, 6

11. In an isosceles triangle the vertex angle (between the two equal sides) is 40°. Each base angle is:
(A) 40° (B) 70°
(C) 80° (D) 100°

12. The angles of a triangle are in the ratio 2 : 3 : 4. The largest angle is:
(A) 40° (B) 60°
(C) 80° (D) 90°

13. An altitude of a triangle is best described as:
(A) Any one of the three sides
(B) A line joining a vertex to the midpoint of the opposite side
(C) The longest side of the triangle
(D) The perpendicular segment from a vertex to the opposite side

14. In a right-angled triangle, the two altitudes drawn from the vertices of the acute angles coincide with:
(A) The two legs (the sides meeting at the right angle)
(B) The hypotenuse and one leg (C) The three medians
(D) Lines that always fall outside the triangle

15. Two sides of a triangle are 8 cm and 5 cm. The third side must lie strictly between:

(A) 5 cm and 8 cm (B) 8 cm and 13 cm

(C) 3 cm and 13 cm (D) 0 cm and 13 cm

16. In ΔABC, ∠A = 2∠B and ∠C = 3∠B. The measure of ∠B is:

(A) 60° (B) 30°

(C) 45° (D) 36°

17. A triangle has exactly two equal angles, each 65°. Classified by its sides it is:

(A) Isosceles (B) Equilateral

(C) Scalene (D) Cannot be decided

18. The exterior angle of a triangle at vertex C equals 130°, and the interior angle at A is 55°. The interior angle at B is:

(A) 65° (B) 50°

(C) 60° (D) 75°

19. Which statement about the angles of a triangle is always TRUE?

(A) A triangle can have two obtuse angles (B) A triangle has at least two acute angles

(C) A triangle can have two right angles (D) Every triangle has exactly one acute angle

20. An equilateral triangle has perimeter 27 cm. Each of its angles measures:

(A) 90° (B) 45°

(C) 60° (D) 27°

21. The three sides of a triangle are 3x, 4x and 5x and its perimeter is 60 cm. The longest side is:

(A) 15 cm (B) 20 cm

(C) 12 cm (D) 25 cm

22. In ΔPQR, the exterior angle at R is 105° and ∠P = 60°. Then ∠Q equals:

(A) 45° (B) 55°

(C) 75° (D) 105°

23. How many of these can be the angles of a single triangle: (40°, 60°, 80°), (90°, 90°, 0°), (30°, 60°, 90°), (50°, 50°, 80°)?

(A) 1 (B) 2

(C) 3 (D) 4

24. A triangle has one angle of 110°. The other two angles:

(A) Are both obtuse (B) Must be acute and add up to 70°

(C) Add up to 110° (D) Can include a right angle

25. The smallest whole-number length (in cm) that can be the third side of a triangle whose other two sides are 6 cm and 9 cm is:

(A) 3 cm (B) 2 cm
(C) 15 cm (D) 4 cm

26. In a triangle the second angle is twice the first and the third is 20° more than the first. The first angle is:

(A) 40° (B) 30°
(C) 50° (D) 35°

27. A triangle has sides 7 cm, 7 cm and 7 cm. Choose the FULLY correct classification:

(A) Equilateral and right (B) Equilateral and acute
(C) Isosceles and obtuse (D) Scalene and acute

28. An exterior angle of a triangle is always:

(A) Equal to the adjacent interior angle (B) Less than each remote interior angle
(C) Equal to the sum of all three interior angles
(D) Equal to the sum of the two remote interior angles

29. The sum of the three exterior angles of any triangle (one at each vertex) is:

(A) 180° (B) 270°
(C) 360° (D) Depends on the triangle

30. In ΔABC, AB = AC and ∠B = 50°. The vertex angle ∠A is:

(A) 80° (B) 50°
(C) 100° (D) 65°

31. Which of these CAN form a triangle?

(A) 1 cm, 2 cm, 3 cm (B) 4 cm, 5 cm, 8 cm
(C) 2 cm, 4 cm, 7 cm (D) 3 cm, 3 cm, 6 cm

32. A right-angled triangle cannot also be:

(A) Isosceles (B) Scalene
(C) A triangle with a 30° angle (D) Obtuse

33. Two angles of a triangle add up to 122°. The third angle is:

(A) 48° (B) 68°
(C) 58° (D) 122°

34. In ΔABC the angles satisfy ∠A : ∠B : ∠C = 1 : 1 : 4. The triangle is:

(A) Obtuse and isosceles (B) Right and isosceles
(C) Acute and equilateral (D) Acute and scalene

35. To construct ΔABC with AB = 4 cm, AC = 5 cm and BC = 6 cm using a compass, you should:

(A) Draw AB = 4 cm, then arc 6 cm from A and arc 5 cm from A
(B) Draw AB = 4 cm, then arc 5 cm from A and arc 6 cm from B; their crossing is C

(C) Draw three separate segments and hope they meet

(D) Draw AB = 4 cm and a perpendicular of 5 cm at A

36. A student claims a triangle with sides 3 cm, 4 cm and 9 cm can be built. The truth is:

(A) Yes — the sides are all different (B) Yes — $4 + 9 > 3$

(C) No — $3 + 4 = 7$ is less than 9, so the short sides cannot meet

(D) No — only equal sides can form a triangle

37. In an isosceles triangle one base angle is 35°. The vertex angle is:

(A) 35° (B) 70°

(C) 145° (D) 110°

38. The exterior angle at one vertex of a triangle is 95°, and the two remote interior angles are equal. Each of them is:

(A) 47.5° (B) 95°

(C) 42.5° (D) 85°

39. A triangle has angles x, x + 10° and x + 20°. The largest angle is:

(A) 50° (B) 60°

(C) 70° (D) 80°

40. Which is impossible for the side lengths (in cm) of a triangle?

(A) 5, 5, 9 (B) 10, 1, 8

(C) 7, 8, 9 (D) 6, 6, 6

41. In ΔABC, side BC is extended to D. If ∠ACD (the exterior angle) = 118° and ∠ABC = 73°, then ∠BAC is:

(A) 45° (B) 35°

(C) 55° (D) 62°

42. A triangle has two sides of length 9 cm and the third side is also a whole number of cm. The largest the third side can be is:

(A) 9 cm (B) 18 cm

(C) 16 cm (D) 17 cm

43. If each base angle of an isosceles triangle is twice the vertex angle, the vertex angle is:

(A) 30° (B) 45°

(C) 36° (D) 72°

44. A triangle is described as having all sides different and all angles less than 90°. It is:

(A) Isosceles and right (B) Scalene and acute

(C) Scalene and obtuse (D) Equilateral and acute

45. The altitude from the right-angle vertex of a right triangle:

(A) Is one of the legs
(B) Coincides with the hypotenuse
(C) Falls outside the triangle
(D) Falls inside, perpendicular to the hypotenuse

46. Two sides of a triangle are 11 cm and 4 cm. The number of whole-number values the third side can take is:

(A) 7
(B) 5
(C) 8
(D) 15

47. In ΔABC, ∠A = 3x, ∠B = 2x and ∠C = x. The triangle is:

(A) Acute
(B) Right-angled
(C) Obtuse
(D) Equilateral

48. An exterior angle of a triangle measures 70°. Then the interior angle adjacent to it is:

(A) 70°
(B) 20°
(C) 110°
(D) 290°

49. A triangle has an angle of 100° and an angle of 50°. Classified by sides, it is:

(A) Equilateral
(B) Isosceles
(C) Right isosceles
(D) Scalene

50. A triangle has angles 35°, 35° and 110°. Classified by its sides AND by its angles it is:

(A) Scalene and obtuse
(B) Isosceles and obtuse
(C) Isosceles and acute
(D) Equilateral and obtuse

51. In an obtuse triangle, the foot of an altitude from a vertex of an acute angle:

(A) Can lie outside the triangle, on the extension of the base
(B) Always lies inside the triangle
(C) Is always the midpoint of the base
(D) Coincides with the obtuse vertex

52. In ΔABC, ∠A = 70° and the exterior angle at B equals 130°. Then ∠C is:

(A) 50°
(B) 70°
(C) 60°
(D) 40°

53. A triangle has sides 5 cm and 5 cm with an included angle of 60°. The triangle is:

(A) Scalene
(B) Right-angled
(C) Obtuse isosceles
(D) Equilateral

54. For three lengths to form a triangle, it is ENOUGH to check that:

(A) The sum of the two largest exceeds the smallest
(B) The sum of the two smallest exceeds the largest
(C) All three are different
(D) Each is less than the sum of the other two squared

55. In ΔPQR, the exterior angle at R is 150° and ∠P = 2∠Q. Then ∠Q equals:

(A) $30°$ | (B) $75°$
(C) $50°$ | (D) $60°$

56. A triangle with angles 89°, 89° and 2° is:

(A) Acute
(B) Obtuse
(C) Right-angled
(D) Not a valid triangle

57. The sides of a triangle are consecutive whole numbers and the perimeter is 12 cm. The shortest side is:

(A) 2 cm
(B) 5 cm
(C) 4 cm
(D) 3 cm

58. In ΔABC, the bisector from A is drawn, and $\angle B = 50°$, $\angle C = 70°$. The angle $\angle BAC$ is:

(A) 50°
(B) 60°
(C) 70°
(D) 120°

59. Two exterior angles of a triangle are 100° and 140°. The third interior angle (at the remaining vertex) is:

(A) 60°
(B) 40°
(C) 80°
(D) 120°

60. A triangle has two sides of length 6 cm and 8 cm, and the third side c is a whole number of cm. How many different whole-number values can c take?

(A) 9
(B) 13
(C) 11
(D) 14

End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 7: A Tale of Three Intersecting Lines

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	B	21	D	31	B	41	A	51	A
2	B	12	C	22	A	32	D	42	D	52	C
3	C	13	D	23	C	33	C	43	C	53	D
4	D	14	A	24	B	34	A	44	B	54	B
5	B	15	C	25	D	35	B	45	D	55	C
6	A	16	B	26	A	36	C	46	A	56	A
7	D	17	A	27	B	37	D	47	B	57	D
8	C	18	D	28	D	38	A	48	C	58	B
9	A	19	B	29	C	39	C	49	D	59	A
10	D	20	C	30	A	40	B	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) A triangle ΔABC has three corner points called vertices (A, B, C), three sides that are segments joining the pairs of vertices (AB, BC, CA), and three angles where the sides meet (∠A, ∠B, ∠C). The other options swap the roles of vertices, sides and angles.

2. (B) Check the triangle inequality on the longest gap: PQ + QR = 4 + 3 = 7, which equals PR = 7, not greater than it. When the two shorter lengths just reach (sum = third), the points lie on one straight line — they are collinear and enclose no area, so no triangle forms.

3. (C) Angle sum = 180°, so third = 180° − (48° + 67°) = 180° − 115° = 65°. The trap 115° is the sum of the two given angles, and 75° comes from a slip.

4. (D) Test the two shortest against the longest: 5 + 8 = 13, which is NOT greater than 13. The inequality must be strict, so no triangle exists. "8 + 13 > 5" is true but irrelevant — you must check the two SHORT sides against the longest.

5. (B) The third side lies strictly between the difference and the sum of the other two: 10 − 7 = 3 and 10 + 7 = 17, so 3 < x < 17. The endpoints are excluded (the ≤ form in (D) is the classic trap; equality gives a degenerate line).

6. (A) All three angles (40°, 60°, 80°) are less than 90°, so the triangle is acute. "Acute" requires ALL angles under 90°, which holds here.

7. (D) The exterior angle equals the sum of the two remote (non-adjacent) interior angles: 120° = 70° + other, so the other = 50°. The trap 70° just repeats the given angle.

8. (C) Two 90° angles already total 180°, leaving 0° for the third — impossible, since all three must sum to exactly 180°. A triangle can have at most one right angle, so this triangle cannot exist.

9. (A) One 90° angle ⇒ right-angled. The two 45° angles are equal, so the two sides opposite them are equal ⇒ isosceles. Hence right-angled AND isosceles.

10. (D) Test each: $2 + 3 = 5 < 6$, so 2, 3, 6 fails the inequality. The others pass ($6+6>6$; $4+6>9$; $5+12>13$), so 2, 3, 6 cannot form a triangle.

11. (B) In an isosceles triangle the two base angles are equal. With vertex angle 40°, the base angles share 180° − 40° = 140°, so each = 70°. The trap 40° wrongly copies the vertex angle.

12. (C) Ratio 2 : 3 : 4 has 9 parts = 180°, so one part = 20°. The largest angle = 4 × 20° = 80°.

13. (D) An altitude is the perpendicular segment from a vertex straight down (at 90°) to the opposite side. Option (B) describes a median, and a side or the longest side is not an altitude in general.

14. (A) In a right triangle, each leg is already perpendicular to the other, so dropping a perpendicular from an acute-angle vertex to the opposite side just lands along the other leg. Thus the two legs ARE the altitudes from the acute vertices.

15. (C) Third side lies strictly between 8 − 5 = 3 and 8 + 5 = 13, i.e. between 3 cm and 13 cm. "0 cm and 13 cm" ignores the lower bound from the difference.

16. (B) $\angle A + \angle B + \angle C = 2\angle B + \angle B + 3\angle B = 6\angle B = 180°$, so $\angle B = 30°$. (Then $\angle A = 60°$, $\angle C = 90°$.)

17. (A) Two equal angles (65° and 65°) mean the two sides opposite them are equal, so the triangle has exactly two equal sides — isosceles. (It is not equilateral, since the third angle is 50°.)

18. (D) Exterior angle at C = sum of the two remote interior angles A and B: $130° = 55° + \angle B$, so $\angle B = 75°$.

19. (B) A triangle can have at most one angle that is 90° or more (two such would already reach 180°), so at least two angles must be acute. Options (A) and (C) are impossible, and (D) is false (e.g. an equilateral triangle has three acute angles).

20. (C) An equilateral triangle has all three angles equal: each = 180°/3 = 60°, regardless of perimeter. The 27 cm is a distractor.

21. (D) $3x + 4x + 5x = 12x = 60$, so $x = 5$ cm. The longest side = $5x = 25$ cm. The trap 15 cm is $3x$ (the shortest side).

22. (A) Exterior angle at R = ∠P + ∠Q, so 105° = 60° + ∠Q, giving ∠Q = 45°.

23. (C) Check each: (40,60,80) sums to 180 ✓; (90,90,0) has a 0° "angle" — invalid; (30,60,90) sums to 180 ✓; (50,50,80) sums to 180 ✓. Three valid sets.

24. (B) The remaining two angles total 180° − 110° = 70°. Since their sum is only 70°, each is under 70° < 90°, so both are acute. They cannot include a right angle (that would need ≥90° but only 70° is left).

25. (D) The third side must satisfy 9 − 6 = 3 < side, so side > 3. The smallest whole number greater than 3 is 4 cm. (3 cm itself is excluded — it would be degenerate.)

26. (A) Let first = x. Then $x + 2x + (x + 20°) = 4x + 20° = 180°$, so $4x = 160°$, $x = 40°$. (Angles: 40°, 80°, 60°.)

27. (B) Equal sides (7, 7, 7) make it equilateral; all angles are 60° < 90°, so it is acute. Hence equilateral and acute. An equilateral triangle can never be right or obtuse.

28. (D) By the exterior angle theorem, an exterior angle of a triangle always equals the sum of the two remote (non-adjacent) interior angles. (It is supplementary to — not equal to — the adjacent interior angle.)

29. (C) At each vertex the interior and exterior angles sum to 180°, so the three exterior angles total 3 × 180° − 180° = 540° − 180° = 360°. The sum of exterior angles of any triangle is always 360°.

30. (A) AB = AC makes the base angles equal, so ∠C = ∠B = 50°. Then the vertex angle ∠A = 180° − 50° − 50° = 80°. The trap 50° wrongly copies a base angle.

31. (B) Test each: 1+2 = 3 not > 3 (fail); 2+4 = 6 < 7 (fail); 4+5 = 9 > 8 ✓; 3+3 = 6 not > 6 (fail). Only 4, 5, 8 works.

32. (D) A right triangle has one 90° angle; a second angle of 90° or more (obtuse) would push the total past 180°. So a right triangle cannot also be obtuse. It can be isosceles (45-45-90) or scalene (30-60-90).

33. (C) Third angle = $180° - 122° = 58°$. The trap 122° repeats the given sum.

34. (A) Ratio 1 : 1 : 4 over 6 parts of 180° gives 30°, 30°, 120°. The 120° angle makes it obtuse; the two equal 30° angles make it isosceles. Hence obtuse and isosceles.

35. (B) Draw the base AB = 4 cm. From A swing an arc of radius 5 cm (= AC); from B swing an arc of radius 6 cm (= BC). Where the arcs cross is C, equidistant correctly from both ends. Option (A) wrongly swings both arcs from A.

36. (C) The two short sides: $3 + 4 = 7$, which is less than 9. They cannot reach across to meet, so no triangle closes. "All sides different" and "$4 + 9 > 3$" are irrelevant to whether it can be built.

37. (D) The two base angles are equal: 35° each. Vertex angle = $180° - (35° + 35°) = 180° - 70° = 110°$. The trap 70° is the sum of the base angles, not the vertex angle.

38. (A) The exterior angle equals the sum of the two equal remote interior angles: $95° = 2x$, so $x = 47.5°$. The trap 42.5° comes from wrongly using 85° (the supplement) and halving.

39. (C) $x + (x + 10°) + (x + 20°) = 3x + 30° = 180°$, so $3x = 150°$, $x = 50°$. The largest angle $= x + 20° = 70°$.

40. (B) Check each against the inequality: $5+5 = 10 > 9$ ✓; $7+8 = 15 > 9$ ✓; for 10, 1, 8 the two short sides $1 + 8 = 9 < 10$ (fail); $6+6 > 6$ ✓. So 10, 1, 8 is impossible.

41. (A) The exterior angle ∠ACD = sum of the two remote interior angles ∠ABC and ∠BAC: $118° = 73° + \angle BAC$, so $\angle BAC = 45°$.

42. (D) With two sides of 9 cm, the third side must be less than $9 + 9 = 18$, so the largest whole number is 17 cm. The trap 18 cm forgets the inequality is strict.

43. (C) Let the vertex angle = v; each base angle = $2v$. Then $2v + 2v + v = 5v = 180°$, so $v = 36°$. (Base angles are 72° each.)

44. (B) "All sides different" ⇒ scalene; "all angles less than 90°" ⇒ acute. So scalene and acute.

45. (D) The altitude from the right-angle vertex is the perpendicular dropped to the hypotenuse; it falls inside the triangle, meeting the hypotenuse at a right angle. It is not a leg (those are the altitudes from the acute vertices).

46. (A) Third side x satisfies $11 - 4 = 7 < x < 11 + 4 = 15$, so $x \in \{8, 9, 10, 11, 12, 13, 14\}$ — that is 7 whole-number values. The trap 5 forgets to count both ends; 15 just lists numbers up to the sum.

47. (B) $3x + 2x + x = 6x = 180°$, so $x = 30°$. Angles are 90°, 60°, 30° — exactly one right angle, so the triangle is right-angled.

48. (C) The interior angle adjacent to an exterior angle is its supplement: $180° - 70° = 110°$. (The remote interior angles, not the adjacent one, sum to 70°.)

49. (D) Third angle = $180° - 100° - 50° = 30°$. The angles 100°, 50°, 30° are all different, so the sides opposite them are all different ⇒ scalene.

50. (B) Two equal angles (35° and 35°) mean the two sides opposite them are equal, so the triangle is isosceles. One angle, 110°, is greater than 90°, so by the angle classification the triangle is obtuse. Hence isosceles and obtuse. It is not scalene (two angles are equal) and not acute (110° exceeds 90°).

51. (A) In an obtuse triangle, the perpendicular dropped from an acute-angle vertex to the (extended) opposite side has its foot OUTSIDE the triangle, on the extension of the base. So it can lie outside — it is not always inside.

52. (C) Exterior angle at B = sum of the two remote interior angles A and C: $130° = 70° + \angle C$, so $\angle C = 60°$.

53. (D) Two sides equal (5, 5) with the included angle 60°: the base angles are equal and share $180° - 60° = 120°$, so each = 60°. All three angles are 60°, making it equilateral (and so the third side is also 5 cm).

54. (B) It is enough to check that the sum of the two SMALLEST sides exceeds the largest — if that holds, every other pair sum (which is even larger) automatically exceeds the remaining side. Checking the two largest against the smallest (A) is automatic and proves nothing.

55. (C) Exterior angle at R = $\angle P + \angle Q$ (remote interior angles). With $\angle P = 2\angle Q$: $150° = 2\angle Q + \angle Q = 3\angle Q$, so $\angle Q = 50°$.

56. (A) All three angles — 89°, 89°, 2° — are less than 90°, so by the angle classification the triangle is acute. (It looks nearly flat, but every angle is still under 90°.)

57. (D) Consecutive sides $n-1$, n, $n+1$ sum to $3n = 12$, so $n = 4$: sides 3, 4, 5. Triangle inequality holds ($3 + 4 > 5$), and the shortest side is 3 cm.

58. (B) $\angle BAC = 180° - \angle B - \angle C = 180° - 50° - 70° = 60°$. The angle bisector mentioned in the stem is a distractor — it does not change $\angle BAC$ itself.

59. (A) The interior angles at those two vertices are $180° - 100° = 80°$ and $180° - 140° = 40°$. The third interior angle = $180° - 80° - 40° = 60°$.

60. (C) The third side c lies strictly between the difference and the sum of the other two: $8 - 6 = 2 < c < 8 + 6 = 14$. So c can be any whole number from 3 up to 13, that is {3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13} — exactly 11 values. The trap 13 counts up to the sum without excluding both endpoints; 14 forgets the inequality is strict.